

ME 451: Refrigeration and Air Conditioning

Course Code	ME 451
Course Name	Refrigeration and Air Conditioning
L-T-P-C	3-0-0-3
Prerequisites	Consent of the faculty member
Intended for	B.Tech.
Distribution	Elective
Approval	3rd Ad-hoc Senate (Nov 2012)

Course Contents

- **Refrigeration**
Introduction to refrigeration system, Methods of refrigeration, Carnot refrigeration cycle, Unit of refrigeration, Refrigeration effect & C.O.P.
- **Air Refrigeration cycle**
Open and closed air refrigeration cycles, Reversed Carnot cycle, Bell Coleman or Reversed Joule air refrigeration cycle, Aircraft refrigeration system, Classification of aircraft refrigeration system. Boot strap refrigeration, Regenerative, Reduced ambient, Dry air rated temperature (DART).
- **Vapour Compression System**
Single stage system, Analysis of vapour compression cycle, Use of T-S and P-H charts, Effect of change in suction and discharge pressures on C.O.P, Effect of sub cooling of condensate & superheating of refrigerant vapour on C.O.P of the cycle, Actual vapour compression refrigeration cycle, Multistage vapour compression system requirement, Removal of flash gas, Intercooling, Different configuration of multistage system, Cascade system.
- **Vapour Absorption system**
Working Principal of vapour absorption refrigeration system, Comparison between absorption & compression systems, Elementary idea of refrigerant absorbent mixtures, Temperature – concentration diagram & Enthalpy – concentration diagram, Adiabatic mixing of two streams, Ammonia – Water vapour absorption system, Lithium-Bromide water vapour absorption system, Comparison.
- **Refrigerants**
Classification of refrigerants, Nomenclature, Desirable properties of refrigerants, Common refrigerants, Secondary refrigerants and CFC free refrigerants
- **Emerging refrigeration Technologies**
Magnetocaloric, electrocaloric, thermoelectric based refrigeration
- **Air Conditioning**
Introduction to air conditioning, Psychometric properties and their definitions, Psychometric chart, Different Psychometric processes, Thermal analysis of human body, Effective temperature and comfort chart, Cooling and heating load calculations, Selection of inside & outside design conditions, Heat transfer through walls & roofs, Infiltration & ventilation, Internal heat gain, Sensible heat factor (SHF), By pass factor, Grand Sensible heat factor (GSHF), Apparatus dew point (ADP)
- **Refrigeration Equipment & Application**
Elementary knowledge of refrigeration & air conditioning equipments e.g. compressors, condensers, evaporators & expansion devices, Air washers, Cooling, towers & humidifying efficiency, Food preservation, Cold storage, Refrigerates Freezers, Ice plant, Water coolers, Elementary knowledge of transmission and distribution of air through ducts and fans, Basic difference between comfort and industrial air conditioning.

Text Books

1. C.P Arora, **Refrigeration and Air conditioning**, Tata McGraw Hill.
2. Manohar Prasad, **Refrigeration and Air Conditioning**, Wiley Eastern Ltd., 1983.
3. Roy. J. Dossat, **Principles of Refrigeration**, Pearson Education 1997